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Courses » Introduction to Internet of Things

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Unit 1 - How to access the portal

Course outline

How to access the portal

- Lesson 1: How to access the home page?
- Lesson 2: How to access the course page?
- Lesson 3: How to access the MCQ, MSQ and Programming assignments?
- Quiz : Assignment 0

Week 1

Week 2

Week 3

Week 4

Week 5

Week 6

Week 7

Week 8

Week 9

Assignment 0

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-07-31, 23:59 IST.

1) Internet of Things (IoT) can be integrated with which of these separate domains: ***1 point***

- a. Cloud-based storage and computing.
- b. Cyber Physical Systems.
- c. Big-data networks.
- d. All of these.

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these.

2) In the current market scenario, IoT captures the maximum share in which one of these? ***1 point***

- a. Industry
- b. Security
- c. Healthcare
- d. Home automation

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Industry

3) The paging operation of a Bluetooth device is used for: ***1 point***

- a. Forming a connection between two Bluetooth devices.
- b. Trying to discover other devices near it.

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4) The main function of the IoT Gateway can be summarized as: ***1 point***

- a. Forwarding packets between LAN and WAN on the IP layer.
- b. Performs application layer functions between IoT nodes and other entities.
- c. Enables local, short-range communication between IoT devices.
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Forwarding packets between LAN and WAN on the IP layer.

5) MANETs are: ***1 point***

- a. Infrastructure oriented
- b. Infrastructure less
- c. Partially infrastructure oriented
- d. Partially infrastructure less

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Infrastructure less

6) The approximation of the measured property sensed by a digital sensor induces: ***1 point***

- a. Quantization error
- b. Hysteresis error
- c. Aliasing error
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Quantization error

7) Nested loop in Arduino is: ***1 point***

- a. Loops in the same function
- b. Loop inside a Loop
- c. Infinte loop
- d. Not supported

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Loop inside a Loop

8) A DC motor is a type of : ***1 point***

- a. Pneumatic actuators
- b. Hydraulic actuators
- c. Electrical actuators
- d. Mechanical actuator

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. Electrical actuators

9) In SDN-based sensor network, one can manage: ***1 point***

- a. Sensor nodes
- b. Sensing delay
- c. Network connectivity
- d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of the above

10) Which of these is NOT related to IoT scalability? ***1 point***

- a. Flexibility within Internet.
- b. Large scale deployment.
- c. Real-time connectivity of billions of devices.
- d. Range of each sensor node.

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. Range of each sensor node.

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Unit 2 - Week 1

Course outline

How to access the portal

Week 1

- Lecture 1: Introduction to IoT- Part-I
- Lecture 2: Introduction to IoT- Part-II
- Lecture 3: Sensing
- Lecture 4: Actuation
- Lecture 5: Basics of IoT Networking- Part- I
- Week 1: Lecture Material
- Quiz : Week 1 : Assignment 1
- Feedback for Week 1

Week 2

Week 3

Week 4

Week 1 : Assignment 1

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-08-15, 23:59 IST.

1) Internet of Things (IoT) can be integrated with which of these separate domains: *1 point*

- a. Cloud-based storage and computing.
- b. Cyber Physical Systems.
- c. Big-data networks.
- d. All of these.

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these.

2) Which of the following is true? *1 point*

- a. IoT is a subset of M2M
- b. M2M is a subset of IoT
- c. IoT is a subset of CPS
- d. CPS is a subset of Web of Things (WoT)

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. M2M is a subset of IoT

3) IoT Proxy operates in: *1 point*

- a. Application layer
- b. Network layer

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4) Which one of these is the first technology in the chronological order of the evolution of IoT? ***1 point***

- a. ATM
- b. Web
- c. Smart City
- d. Smart healthcare

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. ATM

5) The protocol IKEv2 stands for: ***1 point***

- a. Information Key Exchange version 2
- b. Internet Key Exchange version 2
- c. Internet Key Exclusion version 2
- d. Inter-Key Exclusion version 2

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Internet Key Exchange version 2

6) The phenomenon of the output signal slowly changing, independent of the measured property is called: ***1 point***

- a. Aliasing
- b. Drift
- c. Hysteresis
- d. None of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Drift

7) Which of these statements regarding sensors is TRUE? ***1 point***

- a. Sensors are input devices.
- b. Sensors can be analog as well as digital
- c. Sensors respond to some external stimuli.
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

8) Light activated polymers (LAPs) are a special type of: ***1 point***

- a. Light dependent resistors
- b. Shape memory polymers
- c. Solar cells
- d. None of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Shape memory polymers

9) Which of these is a part of the Sensing Layer of the IoT Service Oriented Architecture? ***1 point***

- a. Service integration
- b. Service repository
- c. Business logic
- d. Data sensing and actuation protocols.

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. Data sensing and actuation protocols.

10) A DC motor is a type of : ***1 point***

- a. Pneumatic actuators
- b. Hydraulic actuators
- c. Electrical actuators
- d. Mechanical actuator

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. Electrical actuators

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Unit 3 - Week 2

Course outline

How to access the portal

Week 1

Week 2

- Lecture 6: Basics of IoT Networking-Part II
- Lecture 7: Basics of IoT Networking-Part-III
- Lecture 8: Basics of IoT Networking-Part-IV
- Lecture 9: Connectivity Technologies-Part-I
- Lecture 10: Connectivity Technologies-Part-II
- Week 2: Lecture Material
- Quiz : Week 2 : Assignment 2
- Feedback for Week 2

Week 2 : Assignment 2

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-08-15, 23:59 IST.

1) Which of the following controls the publish-subscribe messaging pattern in MQTT. ***1 point***

- a. MQTT broker
- b. MQTT client
- c. MQTT server
- d. None of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. MQTT broker

2) AMQP is designed for connecting: ***1 point***

- a. Constrained networks
- b. LANs and WANs
- c. Systems and Business processes
- d. None of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. Systems and Business processes

3) The MQTT publish-subscribe protocol is: ***1 point***

- a. Event driven
- b. Broker driven

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4) REST stands for: *1 point*

- a. Representational State Transfer
- b. Resourceful Sensor Transfer
- c. Resourceful State Transport
- d. None of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Representational State Transfer

5) Which of these is TRUE for XMPP? *1 point*

- a. It is decentralized
- b. It supports interoperability
- c. It does not guarantee QoS
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

6) Which of these is a routing protocol for low-power lossy networks over IPv6? *1 point*

- a. RPL
- b. OSPF
- c. Both a and b
- d. None of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. RPL

7) In AMQP, link initiation in order to send and receive messages is done by which frame type? *1 point*

- a. OPEN
- b. ATTACH
- c. BEGIN
- d. FLOW

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. ATTACH

8) RPL supports: *1 point*

- a. Message confidentiality
- b. Loop detection in the routes
- c. Data path validation
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

9) Which statement is TRUE with respect to the IEEE 802.15.4 standard? ***1 point***

- a. It is a low data-rate standard.
- b. Used for architecting wireless PANs
- c. Uses only two layers – PHY and MAC
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

10)In the network layer, Zigbee follows which routing scheme? ***1 point***

- a. AODV
- b. DSDV
- c. RIPv2
- d. OSPF

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. AODV

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Unit 4 - Week 3

Course outline

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Week 1

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- Lecture 11:Connectivity Technologies-Part-III
- Lecture 12: Connectivity Technologies-Part- IV
- Lecture 13: Connectivity Technologies-Part- V
- Lecture 14: Sensor Networks- Part-I
- Lecture 15: Sensor Networks- Part-II
- Week 3: Lecture Material
- Quiz : Week 3 : Assignment 3

Week 3 : Assignment 3

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-09-05, 23:59 IST.

1) The HART protocol was designed for: *1 point*

- a. Home automation networking
- b. Networked Smart Field devices
- c. Vehicular networking
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Networked Smart Field devices

2) Channel hopping is performed at which HART layer? *1 point*

- a. Physical
- b. Data link
- c. Network
- d. Application

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Data link

3) Which amongst these is NOT a NFC mode of communication? *1 point*

- a. Peer to peer
- b. Read/write

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4) Bluetooth is a technology with : ***1 point***

- a. Full duplex links
- b. Half duplex links
- c. Both a and b
- d. None of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Full duplex links

5) The PHY and MAC layers in ISA 100.11A are based upon; ***1 point***

- a. IEEE 802.11a
- b. IEEE 802.15.4
- c. IEEE 802.15.6
- d. IEEE 802.11g

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. IEEE 802.15.4

6) The modulation scheme followed by Z-wave is: ***1 point***

- a. BPSK
- b. QPSK
- c. GFSK
- d. O-QSK

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. GFSK

7) Which of these is NOT a part of sensor web? ***1 point***

- a. Observations & measurements (O&M)
- b. Energy efficiency
- c. Sensor model language (sensorml)
- d. Transducer model language (transducerml or TML)

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Energy efficiency

8) Which topology is supported by Z-wave: ***1 point***

- a. Star
- b. Mesh
- c. Tree
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Mesh

9) Event aware topology management in WSNs include: ***1 point***

- a. Timely detection of an event of interest
- b. Monitoring the event
- c. Adapting with the changes of event state
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

10)WBAN stands for: ***1 point***

- a. Wireless Buffer Area Networks
- b. Wireless Body Area Networks
- c. Wired Body Area Networks
- d. Wired Buffer Area Networks

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Wireless Body Area Networks

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Unit 5 - Week 4

Course outline

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Week 1

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Week 3

Week 4

- Lecture 16: Sensor Networks- Part-III
- Lecture 17: Sensor Networks- Part-IV
- Lecture 18: Sensor Networks- Part-V
- Lecture 19: UAV Networks
- Lecture 20: Machine to Machine Communication
- Week 4: Lecture Material
- Quiz : Week 4 : Assignment 4

Week 4 : Assignment 4

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-09-05, 23:59 IST.

1) With respect to WSN coverage problems, the OGDC algorithm is an acronym for: ***1 point***

- a. Optimal Geographic Density Control
- b. Optimal Geographic Distance Control
- c. Optimal Geographic Distance Coverage
- d. Optimal geographic Density Coverage

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Optimal Geographic Density Control

2) Which of these can be regarded as the salient feature(s) of a 3D UWSN architecture? ***1 point***

- a. Silent & energy-efficient scheme for mobile UWSNs
- b. Iterative approach with less initiators nodes (anchors) required
- c. Mobility prediction
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

3) EM communication for Nano-networks centers around: ***1 point***

- a. 0.1-10 Terahertz channel
- b. 0.1-10 Gigahertz channel

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4) WMSNs stand for: ***1 point***

- a. Wired Multimedia Sensor Networks
- b. Wireless Multimedia Sensor Networks
- c. Wireless Mobile Sensor Networks
- d. Wireless Machine Sensor Network

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Wireless Multimedia Sensor Networks

5) UAV-VANETS link-up may be used for: ***1 point***

- a. Visual guidance
- b. Data-muling
- c. Coverage enhancement
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

6) Sensor coverage in WSNs deal with: ***1 point***

- a. Sensor placement
- b. Density control
- c. Both a and b
- d. None of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. Both a and b

7) In WSN area coverage problem, which is TRUE? ***1 point***

- a. Energy-efficient random coverage
- b. Connected random coverage
- c. A network is connected if any active node can communicate with any other active nodes
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

8) In Human-centric sensing, which of these is NOT a constraint? ***1 point***

- a. Energy of sensing devices
- b. Privacy of users
- c. Participant selection
- d. Network size

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. Network size

9) Which of these topologies cannot be a reliable UAV network topology ***1 point***

- a. Star
- b. Flat mesh
- c. Hierarchical mesh
- d. Tree

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. Tree

10) Low-end M2M nodes are: ***1 point***

- a. Mobile
- b. Energy efficient
- c. Complex
- d. Costly

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Energy efficient

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Unit 6 - Week 5

Course outline

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- Lecture 21 : Interoperability in Internet of Things
- Lecture 22 : Introduction to Arduino- I
- Lecture 23 : Introduction to Arduino- II
- Lecture 24 : Integration of Sensors and Actuators with Arduino- I
- Lecture 25 : Integration of Sensors and Actuators with Arduino- II
- Week 5: Lecture

Week 5 : Assignment 5

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-09-12, 23:59 IST.

1) In what type of pin, data pin of DHT sensor should be connected? ***1 point***

- a. analog pin
- b. digital pin
- c. any one of (a) and (b)
- d. None of this

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. digital pin

2) Why interoperability is required? ***1 point***

- a. Heterogeneous wireless communication protocols
- b. Different programming languages
- c. Different control models
- d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of the above

3) What type of Arduino pin is used to give command to Servo motor? ***1 point***

- a. Analog pin
- b. Digital pin

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4) Arduino boards support output supply voltage of: ***1 point***

- a. 5V only
- b. 3.3 V only
- c. 9 V only
- d. 5V, 3.3V

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. 5V, 3.3V

5) Arduino UNO supports these numbers of PWM pins: ***1 point***

- a. 1
- b. 5
- c. 6
- d. 14

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. 6

6) ***1 point***

```
void Loop(){
        Value = analogRead(LDR);
    if(Value>1000){
            Serial.println(Value);
        }
        delay(100);
        }
```

- a. Prints the value from sensor to console every 1000 miliseconds
- b. Prints the value from the sensor to console every 100 miliseconds
- c. Prints the value from the sensor to console if it is more than 1000, every 100 miliseconds
- d. Prints the value from the sensor to console if it is delayed by 1000 seconds

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. Prints the value from the sensor to console if it is more than 1000, every 100 miliseconds

7) What is the servo library command to move the shaft of Servo motor back and forth across 180 degree? ***1 point***

- a. Knob()
- b. Sweep()
- c. write()
- d. detach()

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. write()

8) Arduino setup requires: ***1 point***

a. Arduino board and IDE

b. Arduino board, programmer board, IDE

c. Both A and B

d. None

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Arduino board and IDE

9) Which of the following library functions can be used to generate a random number in Arduino? ***1 point***

a. randomize()

b. ran()

c. rand()

d. random()

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. random()

10)Which pin of Arduino should be connected with the pin 3 of the 4-pin DHT sensor? ***1 point***

a. Pint 5V

b. Pin 3.3V

c. Pin GND

d. None of the these

No, the answer is incorrect.
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Accepted Answers:
d. None of the these

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Unit 7 - Week 6

Course outline

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Week 6

- Lecture 26: Introduction to Python Programming- I
- Lecture 27: Introduction to Python Programming- II
- Lecture 28: Introduction to Raspberry Pi- I
- Lecture 29: Introduction to Raspberry Pi- II
- Lecture 30: Implementation of IoT with Raspberry Pi- I

Week 6 : Assignment 6

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-09-12, 23:59 IST.

1) Which one is correct for the nested controlling statements in Python? ***1 point***

- a. Elseif
- b. Elif
- c. Ifwhile
- d. none

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Elif

2) Which of the following datatype is correct for the object below? D = ['python', 3.43, 'p', 100] ***1 point***

- a. Array
- b. Dictionary
- c. Tuple
- d. List

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. List

3) Which of the following is mutable object in python? ***1 point***

- a. Tuple
- b. Bool

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4) What is the maximum peripheral current draw allowed in Raspberry pi 3? ***1 point***

- a. 1200mA
- b. 700mA
- c. 500mA
- d. 100mA

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. 1200mA

5) What does the method isatty() return for an existing interactive file stream in Python: ***1 point***

- a. Int
- b. String
- c. Bool:True
- d. Bool:False

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. Bool:True

6) Relay is a kind of: ***1 point***

- a. Sensor
- b. Actuator
- c. Hub
- d. Router

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Actuator

7) What are the mode/modes used for addressing the pins in Raspberry Pi? ***1 point***

- a. GPIO
- b. BCM
- c. BOARD and BCM
- d. GPIO, BCM and CAN

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. BOARD and BCM

8) The BCM 14 pin of Raspberry Pi is: ***1 point***

- a. Physical pin 8
- b. UART
- c. Transmitter pin
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

9) Internet of Things refers to: ***1 point***

- a. Interconnection of devices in a network
- b. Exchange of data between devices in a network
- c. An ecosystem of devices, accessible through internet
- d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of the above

10) Devices in IoT support: ***1 point***

- a. Sensing
- b. Actuating
- c. Data Collection
- d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of the above

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Unit 8 - Week 7

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- Lecture 31 : Implementation of IoT with Raspberry Pi- II
- Lecture 32 : Implementation of IoT with Raspberry Pi- III
- Lecture 33 : Software Defined Networking- Part- I
- Lecture 34 : Software Defined Networking-

Week 7: Assignment 7

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-09-19, 23:59 IST.

1) Opening a file in append mode in Python: ***1 point***

- a. Overrides the existing data in the file
- b. Adds the new data to the end of the existing file
- c. Adds the new data at the beginning of the existing file
- d. Cannot write data to a file in append mode

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Adds the new data to the end of the existing file

2) Control message from a switch to controller in SDN is termed as: ***1 point***

- a. Flow-rule message
- b. PACKET-IN message
- c. Control message
- d. Data message

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. PACKET-IN message

3) Bind() function in Python socket programming is used to: ***1 point***

- a. Specify the port for service on the specified host
- b. Read data from the socket

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4) Protocol used for connectionless socket programming: ***1 point***

- a. TCP
- b. UDP
- c. Both
- d. None

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. UDP

5) When does a switch send PAKCET-IN message to a controller in SDN? ***1 point***

- a. Flow-rule associated with a received flow does not exist
- b. All the time
- c. After deleting a flow-rule
- d. None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Flow-rule associated with a received flow does not exist

6) Which one is an SDN controller: ***1 point***

- a. POX
- b. OpenDayLight
- c. ONOS
- d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of the above

7) Give the output of the following snippet: ***1 point***

```
Name= “Split-this-string”
Name.split(“-”)
```

- a. [“Split”, “this”, “string”]
- b. Split, this, string
- c. [“Split-“, “this-“, “string”]
- d. Split this string

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. [“Split”, “this”, “string”]

8) Application program interface (API) used between application and control planes is known as: ***1 point***

- a. Northbound API
- b. Eastbound API

c. Southbound API

d. Westbound API

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Northbound API

9) An SDN Controller can be placed based on: ***1 point***

a. Hierarchical manner

b. Star-topology

c. Mesh topology

d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of the above

10)Which of the following is not a function in matplotlib library of Python: ***1 point***

a. Display()

b. Show()

c. Subplot()

d. Title()

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Display()

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Unit 9 - Week 8

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- Lecture 36 : Software Defined IoT Networking- II
- Lecture 37 : Cloud Computing- Fundamental
- Lecture 38 : Cloud Computing- Service Model
- Lecture 39 : Cloud Computing- Service

Week 8 : Assignment 8

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-09-26, 23:59 IST.

1) The major challenge(s) in Software Defined IoT is(are): ***1 point***

- a. Mobility of devices
- b. Heterogeneity of devices
- c. Resource-constraint
- d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of the above

2) From the below given options, which is not a characteristic of cloud computing? ***1 point***

- a. Low management effort
- b. Low level generalization of computation
- c. On-demand network access
- d. Configurable computing resources

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Low level generalization of computation

3) Pre-configure facility for allocation of virtualized resources comes under ***1 point***

- a. Availability and reliability
- b. Scalability and elasticity

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4) If the devices are highly mobile in IoT, one of the disadvantage of using SDN can be: ***1 point***

- a. High control overhead
- b. Low data rate
- c. All of the above
- d. None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. High control overhead

5) Broad network access in cloud deals with ***1 point***

- a. Dynamic allocation for scale out
- b. Recorded resource usage
- c. Availability of cloud resources over the network
- d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. Availability of cloud resources over the network

6) Mice flows have: ***1 point***

- a. Large duration
- b. Short duration
- c. Moderately large duration
- d. None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Short duration

7) What is the full form of HDFS? ***1 point***

- a. Hadoop distributed file system
- b. Hadoop database file system
- c. Half-duplex file system
- d. Hadoop decentralized file system

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Hadoop distributed file system

8) Which of the following is not a part of Hadoop? ***1 point***

- a. HDFS
- b. YARN
- c. Both the above
- d. None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. None of the above

9) GreenCloud Simulator is designed based on ***1 point***

- a. NS-3
- b. NS-2
- c. CloudSim
- d. DCSim

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. NS-2

10) The disadvantage(s) of private cloud is/are: ***1 point***

- a. Total control over the system and data
- b. Regular maintenance
- c. Minimum security concerns
- d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Regular maintenance

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- Lecture 41 : Cloud Computing- Practical
- Lecture 42 : Sensor-Cloud- I
- Lecture 43 : Sensor-Cloud- II
- Lecture 44 : Fog Computing- I
- Lecture 45 :

Week 9 : Assignment 9

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-10-03, 23:59 IST.

1) Resource usage information in service accounting is obtained in the form of *1 point*

- a. Data set
- b. Tables
- c. Records
- d. Any form

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. Records

2) We normally take snapshot in Openstack, _____. *1 point*

- a. After the installation of an instance
- b. Before the installation of an instance
- c. It is an inbuilt image in Openstack
- d. It should be downloaded from the internet

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. After the installation of an instance

3) Sensor-Cloud deals with *1 point*

- a. Sensor-for-Service
- b. Sensor-as-a-Service

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4) Which of the following statement is TRUE in sensor-cloud *1 point*

- a. End user procures sensors and lends these to sensor-cloud service provider
- b. Sensor-cloud is based only on dumping the sensor data to the cloud
- c. Single sensor nodes is shared dedicatedly with only one end user
- d. None of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. None of these

5) Blue Pill attack on hypervisor is a threat example of: *1 point*

- a. Host security at PaaS level
- b. Host security at SaaS level
- c. Host security at IaaS level
- d. None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. Host security at IaaS level

6) Why data volume is a problem in IoT-based cloud computing? *1 point*

- a. because data are encrypted and hard to analyze
- b. because data coming from IoT devices are always in raw format and difficult to store
- c. because the density of IoT devices is increasing each day and in turn the volume of generated data is growing very fast
- d. because IoT device density is decreasing each day and in turn the volume of generated data are decreasing very fast

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. because the density of IoT devices is increasing each day and in turn the volume of generated data is growing very fast

7) Which layer is responsible for user authentication in cloud computing *1 point*

- a. IaaS layer
- b. PaaS layer
- c. SaaS layer
- d. Both IaaS and PaaS layer

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. PaaS layer

8) What is a good approach to reduce latency? *1 point*

- a. Analyzing data close to the data source
- b. Analyzing data in cloud

c. Increasing number of nodes

d. Decreasing number of nodes

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Analyzing data close to the data source

9) What are less time-sensitive data? ***1 point***

a. Data which can be analyzed after seconds or minutes

b. Data which should be analyzed within fraction of a second

c. Data which can be wait for hours, days, weeks

d. Data which can be wait for months

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Data which can be analyzed after seconds or minutes

10) How many layers exist in CloudSim architecture? ***1 point***

a. 4

b. 1

c. 3

d. 2

No, the answer is incorrect.
Score: 0

Accepted Answers:
c. 3

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- Lecture 56: Data Handling and Analytics-II
- Lecture 57: Case Study: Agriculture
- Lecture 58:

Week 12 : Assignment 12

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2018-10-24, 23:59 IST.

1) Passive and omnidirectional sensor is *1 point*

- a. Temperature
- b. Water-level
- c. Soil moisture
- d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of the above

2) If all the data points are same, statistical dispersion becomes – *1 point*

- a. 1
- b. 10
- c. 100
- d. None

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. None

3) Which of the following can be the reasons for packet drop in IoT network? *1 point*

- a. Dense foliage
- b. High temperature

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Case Study: Activity Monitoring - II

Week 12 : Lecture Material

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4) The characteristics of a sensor node are ***1 point***

- a. Energy-efficiency
- b. Distributed sensing
- c. Low-cost
- d. All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of the above

5) What is the network topology of Ambusens WBAN devices ***1 point***

- a. Bus topology
- b. Star Topology
- c. Mesh Topology
- d. None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
b. Star Topology

6) Long term trend tracking and preventive care in IoT healthcare is possible due to ***1 point***

- a. ability of smart devices to collect data autonomously
- b. ubiquitous connectivity
- c. smart algorithms and analytics
- d. all of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. all of the above

7) IoT-based sensing layer devices typically communicate with each other using which of the following? ***1 point***

- a. Short range communication technologies such as Bluetooth and Zigbee
- b. Long range communication technologies such as WLAN, WiMAX and LTE
- c. Both may be used
- d. Devices do not communicate with each other

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Short range communication technologies such as Bluetooth and Zigbee

8) Activity monitoring in smart homes: ***1 point***

- a. Provides better quality of life
- b. Provides information accurately in a reliable manner
- c. Provides continuous monitoring support.
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

9) Which of these sensors can be most cheaply used for activity monitoring in wearables? ***1 point***

- a. Accelerometer
- b. Cameras
- c. LIDARs
- d. LED

No, the answer is incorrect.
Score: 0

Accepted Answers:
a. Accelerometer

10)Network-based activity recognition is: ***1 point***

- a. Larger and processing intensive
- b. Low power consumption
- c. Dependent on network connection
- d. All of these

No, the answer is incorrect.
Score: 0

Accepted Answers:
d. All of these

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